AI Semiconductor Supercycle Risk Brief

Tail Risk & Volatility Persistence (2018–2023): Evidence from Student-t GARCH(1,1)

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Deliverable: 2-page brief + reproducible code (https://github.com/siruizhu773-source/AI-Semiconductor)

Business Question

How do tail risk and volatility regimes differ across AI-semiconductor assets, and which model best captures extreme downside moves?

Asset Universe

NVDA (AI infrastructure leader), INTC (legacy incumbent under transition), TSM (geopolitical/supply-chain bellwether) and Nasdaq Composite (^IXIC) as the tech market benchmark.

Key Findings

・Fat tails are real
Non-normal returns imply Gaussian risk metrics underestimate extreme moves.

・Volatility shocks persist
For NVDA, volatility persistence is near unity ($\alpha+\beta \approx 0.974$) suggests volatility shocks remain elevated for longer.

・ Student-t GARCH outperforms Normal GARCH
NVDA's AIC improves from 7610.66 (Normal) to 7496.27 (Student-t), with $\nu \approx 6.07$ supporting heavy-tails risk.

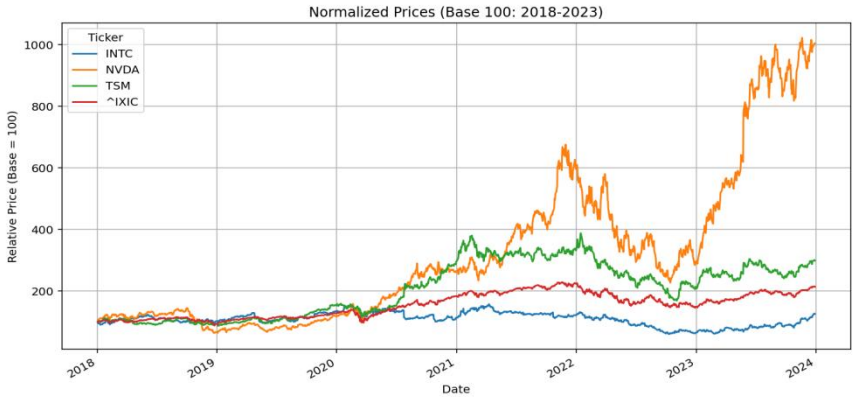

Figure 1. Normalized Prices (Base = 100: 2018 - 2023)

Figure 1 Caption：
Normalized performance highlights divergent regimes and asset-specific behavior within the same sector, with ^IXIC serving as the systemic benchmark to separate market-wide vs idiosyncratic risk.

Figure 2 Caption： Histogram + density curves visually demonstrate leptokurtic (fat tailed) return behavior, motivating the switch from Gaussian risk assumptions to Student-t modeling.

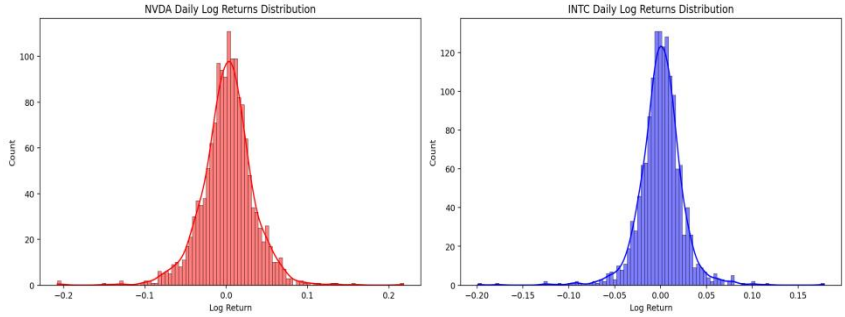

Figure 2. Evidence of Fat Tails (NVDA vs INTC Return Distributions)

Methodology, Model Diagnostics & Actionable Recommendations (Data Analytics Focus)

Method

Data Pipeline (Daily, 2018–2023)

- Data source: daily close prices via yfinance for NVDA / INTC / TSM / ^IXIC.
- Quality check: missing values validated; clean time series used for modeling.
- Return transformation: log returns for time-additivity and modeling suitability.
- Scaling: returns ×100 for numerical stability in GARCH maximum likelihood estimation

Why this window (2018–2023)?

Chosen to capture a complete macro/tech "super-cycle" including trade tensions (2018–2019), COVID shock and easing (2020–2021), aggressive tightening (2022), and the generative-AI breakout (2023).

Daily frequency yields 1,500 observations, improving robustness vs weekly/monthly alternatives.

Modeling Framework

- EDA --- confirm non-normality (skewness, excess kurtosis)
- Engle's ARCH LM test --- verify volatility clustering
- Fit GARCH(1,1): Normal vs Student-t --- compare AIC
- Diagnostics: standardized residuals and ACF of squared residuals (white-noise check)

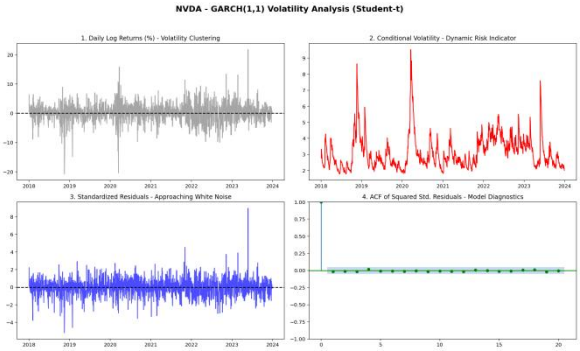

Figure 3. GARCH Diagnostics Dashboard（NVDA）

Caption： Conditional volatility (red line) reacts sharply to macro shocks and remains elevated, consistent with high persistence; standardized residual diagnostics support model adequacy.

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Step 1 & 2: Model Selection (Normal GARCH vs. Student-t GARCH)
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Objective: Empirically justify the necessity of fat-tailed risk assumptions.


===  Model Selection Summary ===
Ticker  Normal GARCH (AIC)  Student-t GARCH (AIC) Winning Model  AIC Reduction
  NVDA            7610.66               7496.27     Student-t         114.39
  INTC            6759.14               6493.33     Student-t         265.81
   TSM            6425.09               6363.59     Student-t          61.50
 ^IXIC            5053.36               5005.12     Student-t          48.23

Conclusion: The Student-t GARCH significantly outperforms the Normal GARCH across all assets.
Proceeding with the Student-t distribution for robust tail-risk analysis.
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Figure 4. Model Selection: Normal GARCH vs Student-t GARCH (AIC Comparison)

Caption ： Student - t specifications outperform Normal GARCH across the portfolio, empirically justifying fat tailed risk assumptions under AI-cycle volatility regimes.

Actionable Recommendations

Upgrade risk metrics

significant fat tails, replace Gaussian VaR with Student - t Expected Shortfall (CVaR) for downside risk governance in AI cycle assets.

Hedge with duration matched to persistence

Because volatility persistence is very high ($\alpha + \beta$ approaching 1), risk spikes do not fade quickly. Use long - dated hedges rather than short term protection when regimes shift.

Portfolio construction implication

High idiosyncratic volatility in NVDA - type assets requires stronger counter - balancing with lower - beta or defensive exposures to stabilize drawdowns during tech sell - offs